

**Central Analytical Laboratories**

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**CERTIFICATE OF ANALYSIS**

January 10, 2020

**Client:** Standard Process  
1200 W. Royal Lee Drive  
Palmyra, WI 53156 United States

**COMMODITY:** Hemp Seed and Stalk Oil

**LOT NUMBER:** PO000065134-2-1

**QA18L: Product Specification Review**

1. The hemp seed and stalk oil (PO000065134-2-1) meets the requirements for pesticides as defined by USP 561.
2. The hemp seed and stalk oil (PO000065134-2-1) meets the requirements for heavy metals as defined by USP 2232.
3. The hemp seed and stalk oil (PO000065134-2-1) meets the requirements for Microbiological levels as defined by USP 2023 (Botanical Extracts).



Victoria Siegel, Analytical Service Manager

ECAL ID: 19-12130026

# CERTIFICATE OF ANALYSIS



12661 HOOVER STREET GARDEN GROVE, CA 92841 | P. 714-754-4372 | F. 714-668-9972 | WWW.ALKEMIST.COM

**Report Issued To:** Standard Process, Inc  
1200 West Royal Lee Drive  
Palmyra WI 53156

**Sample Name:** Hemp Stalk Oil  
**Description:** Oil; Dark Oil  
**Lot #:** PO000065134-2-1  
**AL #:** 19155UGH\_1  
**Analysis ID:** 120448  
**Received:** 06/04/19

## Determination of Cannabinoids Content by UPLC

| Ret. Time (min) | Compound Name                             | Prep 1 (%) | Prep 2 (%) | Average (%) | Specification | Result |
|-----------------|-------------------------------------------|------------|------------|-------------|---------------|--------|
| 7.1             | $\Delta^9$ -Tetrahydrocannabinidiol (THC) | 0.104      | 0.104      | 0.104       | NMT 0.2%      | Pass   |

### Chromatographic Conditions:

Method: ATM-815-0229  
Column: AP213 Xbridge BEH C18 2.5 $\mu$ m (3.0 x 100 mm)  
Temperature: 30°C  
Flow Rate: 0.4 mL/min  
Injection Volume: 1  $\mu$ L  
UV Detection: 210 nm  
Mobile Phase: 0.1% Formic Acid in 10% Acetonitrile  
0.1% Formic Acid in 90% Acetonitrile  
HPLC Instrument: UPLC\_1

### Sample Preparation:

Mixed sample well. Transferred approximately 50 mg of sample to a 10 mL volumetric flask. Filled to volume with methanol. Vortexed for 30 seconds and sonicated for 15 minutes at room temperature. Filtered a portion through a 0.2  $\mu$ m PTFE syringe filter into an HPLC vial for analysis.

### Report Summary:

Conclusion: The "Hemp Stalk Oil" sample contains 0.1% THC on the as-is basis.  
OOS Reference: N/A  
Notebook Reference: LC139 p. 35  
Revision 1: Removed chromatogram



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Adobe Acrobat version:  
2019.021.20061

**Analysis Date : 06/07/19**  
**Rev 1: 12/18/19 CD**

**Analyzed By: C Deneuve**

**Authorized By: Colin Gray,**  
**Lead Analytical Chemist**

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